

Approximating spatial processes with too many knots degrades predictive accuracy of SPDE models

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Abstract

1. Spatial and spatiotemporal models are seeing increasingly used in ecology and related fields to model latent distributions. Applications relying on these methods have used them to track population change over time, assess changing species distributions, and model spatial population processes (predation, habitat use, fishing impacts, etc).
2. Computational advances over the last decade have allowed for the rapid estimation of spatial approximations to high dimensional spatial surfaces (100s or more georeferenced points), and these methods have been accessible in software implementing the Integrated Nested Laplace Approximation (INLA). One of the most critical decisions for building INLA-based models is how to construct the spatial approximation (“mesh”). Little guidance has been established in constructing these approximations, with many previous applications suggesting that higher resolution meshes result in better performance. Using a publicly available dataset on commercially fished species on the west coast of the USA, we develop spatial models of species occurrence and catch rates for 4 species. We conduct a series of sensitivity analyses to understand how mesh resolution affects the predictive accuracy of models predicting species occurrences and catch rates.
3. Our results highlight that for both presence-absence (binomial) and positive (Gamma) models, the

finest scale spatial approximations considered overfit the data and resulted in decreased out of sample predictive performance. For each of the species and models considered, there appeared to be an optimal range of vertices (knots) used in constructing the spatial approximation. These results held across a range of spatial cross-validation sampling schemes, including folds selected in 1-dimensional latitudinal blocks, and the use of 2-dimensional spatial blocks.

4. Results from our work highlight that practitioners should not assume that the performance of INLA-based models gets better with higher resolution. Instead, each dataset likely has a range of resolutions that is optimal, and these optimal ranges may be identified using spatial cross-validation approaches, such as the sampling procedures used in our analysis. Selecting models with the appropriate spatial complexity is expected to improve parameter estimates and derived quantities (estimates of population abundance, distribution, etc.).

Key words

Spatial, spatiotemporal, Integrated Nested Laplace Approximation (INLA), Gaussian random field, mesh, predictive accuracy

Introduction

Recent computational advances have allowed for the rapid estimation of large spatial and spatiotemporal models in ecology and related fields (Banerjee & Fuentes 2012). Motivating questions for these applications to large georeferenced datasets including tracking abundance in time and space, identifying environmental predictors of occurrence, or quantifying range shifts in distribution (Ross *et al.* 2012; Miller *et al.* 2013; Thorson & Barnett 2017).

There are a number of statistical approaches for developing models with spatial components; examples include generalized additive models (GAMs) (Murase *et al.* 2009; Forney *et al.* 2012), non-parametric approaches (Mutanga *et al.* 2012), and applications of Gaussian random fields (Latimer *et al.* 2009; Datta *et al.* 2016). Advances by Rue *et al.* (2009) have enabled the development of the Integrated Nested Laplace Approximation (INLA) as a fast and efficient approximation to estimating high-dimensional Gaussian random fields. These models have been extended to include the ability to fit geographic barriers, such as islands or lakes, anisotropic covariance functions, and non-Gaussian data (Bakka *et al.* 2018). In addition to estimating Bayesian posterior distributions (Lindgren & Rue 2015), INLA has been integrated with Template Model Builder (TMB; (Kristensen *et al.* 2016; Osgood-Zimmerman & Wakefield 2021)) to enable auto-differentiation and has been adopted in many spatial fisheries applications around the world (Anderson

et al. 2019a; Thorson 2019b; e.g., Johnson *et al.* 2019).

Constructing spatiotemporal models with INLA or related software involves many of the same considerations as with simpler non-spatial models. For example, modelers frequently ask: “Which covariates are most important?” and “Does my model do a satisfactory job of making predictions?” Spatial or spatiotemporal models may include additional considerations, such as specifying the form of spatiotemporal variation (Anderson & Ward 2019). One of the more complicated and important decisions—particularly for INLA based models—lies in specifying the dimensionality of the latent spatial fields (Lindgren & Rue 2015). Unlike Generalized Additive Models (GAMs), which often penalize smooth terms (Wood 2003), penalties in overfitting INLA models are not as apparent.

One of the first steps in fitting a model with INLA involves choosing the appropriate resolution of the spatial field using a “mesh” (a map consisting of small triangles, e.g., Figure 1). By estimating values of the spatial field at the vertices (or “knots”), the spatial field may be triangulated to coordinates where data are observed or predictions are made (Lindgren *et al.* 2011). There are a number of decision points to be made in constructing the mesh, including whether to include a boundary, whether the boundary is regular, the maximum edge of any triangle (in the core or boundary area), and the minimum angles allowed by the triangles (Gómez-Rubio). Each of these decisions can impact the density and number of small triangles that make up the mesh—and in turn, the accuracy of predictions. Though the mesh construction is important, this step has been omitted from best practices guides using an INLA-based approach (Thorson 2019a), and we are only aware of one sensitivity analysis examining the importance of mesh construction (Righetto *et al.* 2020). While many previous applications and guidance around INLA have suggested that increasing the resolution of the mesh improves predictive accuracy (Thorson 2019a; Godana *et al.* 2019; Wilson 2021; Williamson *et al.*), a much smaller number of studies have noted that finer meshes don’t always translate into better performance (Huang *et al.* 2017).

The objective of this paper is to examine whether higher resolution spatial approximations in INLA translate to better predictive accuracy. Using a series of analyses on a real-world case study of groundfishes in the Northeast Pacific Ocean, we evaluate the out of sample predictive performance as a function of mesh resolution (number of knots). We further evaluate these results across a range of cross-validation sampling schemes (number of spatial blocks and folds), including an extreme case of leave-one-out cross validation (LOOCV). All data for our analysis are publicly available, and code and data to reproduce our analysis is available on Github (XXXX) and Zenodo (on publication).

Methods

Data

As a case study to evaluate the effect of mesh resolution on predictive ability of INLA-based models, we use a publicly available dataset of catch rates from scientific surveys in the Northeast Pacific Ocean (<https://www.nwfsc.noaa.gov/data>). The West Coast Bottom Trawl Survey (WCBTS) dataset represents a fishery-independent survey collected annually over a 19-year period, 2003-2021, covering the California Current ecosystem on the west coast of the USA (Keller *et al.* 2017). The survey is designed to estimate the abundance, sizes, and age composition of commercially and recreationally targeted groundfish species found in near-bottom habitats. The fundamental methods of the survey, sampling intensity, gear, seasonal and spatial coverage, and random stratified design have remained relatively constant. Because of the large number of model runs needed, we restricted our analysis to only use data from 2018 ($n = 702$ observations). Though hundreds of species were encountered in the survey, we restricted our analysis to four species that are of interest to fisheries, but also represent a range of occurrences: Dover Sole (*Microstomus pacificus*, encountered in 593 tows), Petrale Sole (*Eopsetta jordani*, 291 tows), Lingcod (*Ophiodon elongatus*, 225 tows), and Darkblotched Rockfish (*Sebastes crameri*, 105 tows). The overall mean distance between tow locations in the WCBTS was 676.7km ($\tilde{x} = 603.1\text{km}$).

Modeling

Because of the presence of zeros and positive continuous values in fish survey datasets such as the WCBTS data, conventional approaches to analyzing these data have used a delta or hurdle model to separately model occurrence and positive catches (Thorson 2019a; e.g., Johnson *et al.* 2019). We adopt a similar approach in our modeling, using a binomial distribution with a logit link to model species occurrence, and a Gamma distribution with a log link to model positive catch rates (kg per km²). For simplicity and consistency with fisheries assessments using these data (@ Johnson *et al.* 2019), we fit models estimating an intercept and spatial field throughout, e.g.

$$g(u_s) = B_0 + \omega_s$$

where ω_s is the estimated spatial field, B_0 is the intercept, u_s is the expected value for point in space s , and $g()$ is an inverse link function.

All of the meshes in our analysis used a non-convex boundary and offset based on 5% of the data

(Figure 1). We fixed the maximum edge of triangles in the mesh at 100km in the core area and 500km closer to the boundary (Figure 1). Throughout our analyses, we adjusted the mesh resolution by varying the cutoff distance (representing the distance where two points are treated as a single vertex) because the cutoff distance has been shown to have the greatest impact on measures of model performance from mesh construction (Righetto *et al.* 2020). We fit all models using R-INLA (Lindgren & Rue 2015; R Core Team 2021) and R 4.1.0 (R Core Team 2021) and specifically the ‘inlabru’ R package, which is a convenient wrapper for the INLA software (Bachl *et al.* 2019). We used penalized complexity priors on the Matérn covariance parameters (Simpson *et al.* 2015; Osgood-Zimmerman & Wakefield 2021). Specifically, priors on the Matérn spatial range (κ) and marginal standard deviation (σ_ω) are constructed as a function of the tail probabilities (Simpson *et al.* 2017; Fuglstad *et al.* 2018). We assigned priors so that $Pr(\kappa < 20) = 0.05$ and $Pr(\sigma_\omega > 5) = 0.05$. TODO (AND OTHER PRIORS).

Sensitivities

As a preliminary evaluation of whether the spatial resolution of INLA-based models can result in overfitting, we evaluated 20 cutoff values for each species and hurdle sub-model included in our analysis (cutoff values ranging from 3–120km). There is an inverse relationship between cutoff values and mesh resolution or knots; while the meshes varied slightly by species, this range of cutoff produced meshed that ranged in size from 75–1348 knots. High-resolution meshes with more knots than observations were included as an upper extreme, though many applications of INLA to similar survey data have used 500–600 knots (e.g., Thorson 2019a; Tolimieri *et al.* 2020).

To evaluate model performance, we conducted 10-fold cross-validation. Though not a 1-dimensional dataset, the strong latitudinal gradient has driven spatial stratification in previous analyses (Keller *et al.* 2017) (Figure 1). We a priori divided the survey domain into 10 spatial blocks by latitude quantiles with approximately the same number of points in each box (the latitudinal mean distance of each box was 178.1km, $\sigma=56.2$). Each model was run using inlabru (@ Bachl *et al.* 2019), and the fit to each set of training data was used to predict observations in the held out block. The predictive density was calculated using the binomial or Gamma densities, and total predictive densities were calculated by summing across folds. To contrast the out of sample predictive performance with fit to the training data, we also calculated the predictive density of the training data, separately for each fold.

As a second sensitivity, we evaluated whether the 1-dimensional cross-validation sampling may influence inference about predictive ability. Recent work has demonstrated that for spatially structured data, creating random folds that ignore spatial autocorrelation may bias results (Valavi *et al.* 2019). Using the

presence-absence data from Dover Sole, we examined the cross validation performance across a range of mesh resolutions (cutoff values ranging from 3–120, or 75–1348 knots), number of spatial blocks (19–218) and number of folds used for cross validation (10, 40). Selection of spatial blocks was done using the ‘blockCV’ package in R (Valavi *et al.* 2019) with varying distance bands between test and training sets (25–150km). These bands are used to create buffers around test data, with larger bands creating more separation between test and training data.

To further validate the inference from the spatial cross-validation, we examined the performance of alternative measures of out of sample predictive performance. We focused on the conditional predictive ordinate (CPO, (Pettit 1990)), which is an approximation to the leave-one-out cross-validation score (LOOCV). Ideally, we would use cross validation to evaluate the out of sample predictive accuracy of a model. The total predictive log density can be represented as the sum of individual log posterior probabilities, $\sum_{i=1}^n \log[P(Y_i|Y_{\neq i})]$, where $P(Y_i|Y_{\neq i})$ is the probability of test data i . Because conducting true leave-one-out cross-validation may be prohibitive for large datasets, INLA uses CPO to approximate the out of sample posterior probability for each observation. Previous research has shown that the CPO approximation performs similarly to MCMC cross-validation (Held *et al.* 2010), yet it is less clear whether CPO is similar to true cross validation in comparing mesh resolution. Focusing on the presence-absence model of Dover Sole (Figure 2), we first calculated the total CPO score across a range of cutoff values. For comparison, we then calculated the true predictive density using LOOCV across a similar range of cutoffs (fitting the INLA model 702 times for each value of the range).

As a final sensitivity, we compared parameter estimation using INLA to estimation of the same models using TMB (Kristensen *et al.* 2016). Specifically, we performed estimation using the R package ‘sdmTMB’ (Anderson *et al.* 2019b), which uses the INLA software to create meshes, but takes advantage of autodifferentiation in TMB for faster parameter estimation.

Results

Our initial results fitting the binomial and Gamma models across a range of species and mesh cutoff values demonstrated that after some optimal spatial cutoff of triangles in the mesh, predictive accuracy of the spatial models deteriorated as mesh resolution was increased. Binomial models of Darkblotched Rockfish (Fig. 2) and Gamma models for all species also supported a decline in performance with the coarsest mesh resolutions (Fig. S1). The binomial sub-model suggested that the optimal number of knots was 200–250 (Fig. 2), and the positive Gamma sub-model indicated even smaller knot values had nearly equivalent out of sample predictive performance (Fig. S1). In contrast to the decline in out of sample performance with high

mesh resolution, we found the opposite to be true for in sample data; for both the binomial and Gamma models, predictive performance generally increased with the number of knots (Fig. 2, Fig. S1).

To verify that our results are not driven by the choice of sampling scheme for spatial cross validation, we re-fit the binomial model for Dover Sole across a range of spatial sampling blocks (Fig. 3) and folds for cross-validation (10 and 40, Fig. S3). For relatively small numbers of spatial blocks (19, 25) there appears to be a decline in predictive performance at small numbers of knots (Fig. 3, Fig. S3), but with large numbers of spatial blocks (> 50), coarse resolution meshes have similar performance as models with moderate numbers of knots.

Because k-fold cross validation requires significant computation time, approximations like CPO have been used as a measure of predictive accuracy to LOOCV. Though conventionally used to compare models with alternative fixed effects, we compared CPO approximations to LOOCV, as an extreme case of cross-validation. Both CPO and LOOCV suggested that too many knots can result in diminished predictive accuracy (Figure 4). The biggest difference in these metrics was that the LOOCV suggested that more than 400 knots resulted in decreased performance, while CPO indicated a performance decline after 675 knots (Figure 4).

Finally, we examined the effects of increasing mesh resolution on estimated model parameters. We found little evidence of patterns between mesh resolution, estimated Matérn parameters (κ , σ_ω), and predictive log densities. While the estimated range (κ) generally declined with large numbers of knots (Fig. 5), there are less obvious patterns in the marginal standard deviation σ_ω (Fig. 5). The estimated fixed effect B_0 representing the mean of the spatial field generally declined with increasing mesh resolution (Figure S2), because as the mesh increases in complexity and explains more variation in observed data, the intercept is expected to shrink toward the mean.

TODO (WORK IN SDMTMB COMPARISON)

Discussion

1. Larger meshes aren't necessarily better. Others have pointed this out, (Huang *et al.* 2017), but it gets overlooked

Datasets like the WCTBS support an optimal mesh of ~ 300 or fewer knots

2. Selecting the location and number of knots isn't unique to INLA based models – other examples include GP models (Banerjee *et al.* 2012; Garton *et al.* 2020) or P-splines (Ruppert 2002)

204 3. Rules of thumb? This is tricky because it's somewhat dataset specific. The blockCV MEE paper
205 suggests defining spatial blocks based on residual spatial autocorrelation – which involves fitting the
206 model, so somewhat circular [and tools like calculating the variogram are made difficult by non-normal
207 response data]. Most advanced GP approaches include knot selection as a parameter, or do an adaptive
208 grid search

209 The number of knots needed to generate a model with high predictive accuracy depends on the other
210 features of the model. Some of this is based on data (e.g. homogenous spp like Dover sole vs patchily
211 distributed rockfish), but a lot of this is dependent on how much variation is parsed into observation error.
212 High obs errors will result in flatter spatial fields, which will require fewer knots.

213 Cross-validation is important for identifying the range of knots with highest predictive accuracy

214 4. Make bridge with GAMs. GAMs allow for a flexible range of smoothing functions, including gaussian
215 processes with the Matern covariance. (Yue *et al.* 2014; Miller *et al.* 2020) did a great job linking
216 these and showing the common interpretations between SPDEs and smoothing splines. One option
217 would be to use GAMs to compare alternative meshes / numbers of knots, but that begs the question
218 why we aren't just doing this in mgcv in the first place.

219 5. Sub-optimal meshes are likely to have an impact on derived things we care about, like indices of
220 abundance – both bias and precision, fig s2

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223 our analysis and presentation of results.

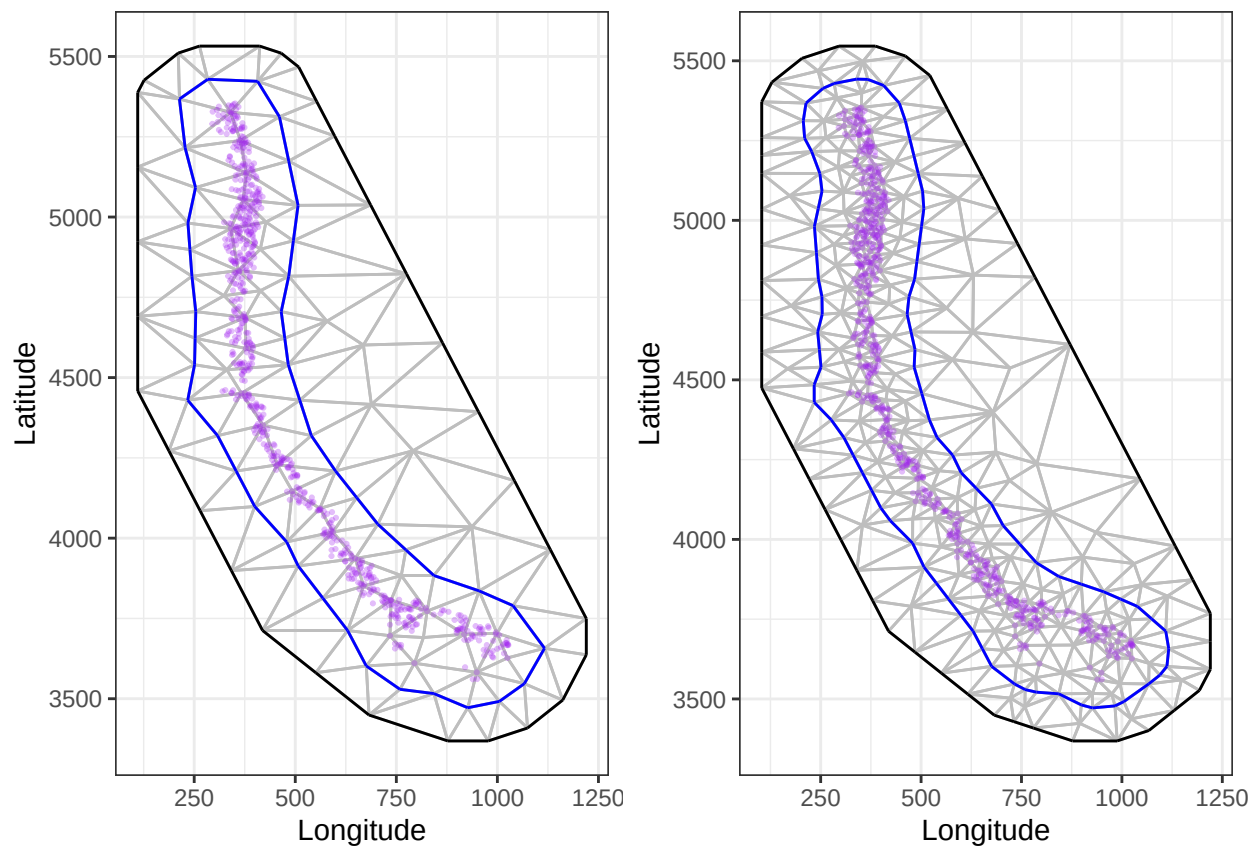


Figure 1: Example of two meshes constructed for the California Current Ecosystem. Cutoff values of 80 and 20km are used, translating into 110 and 394 knots (vertices) respectively.

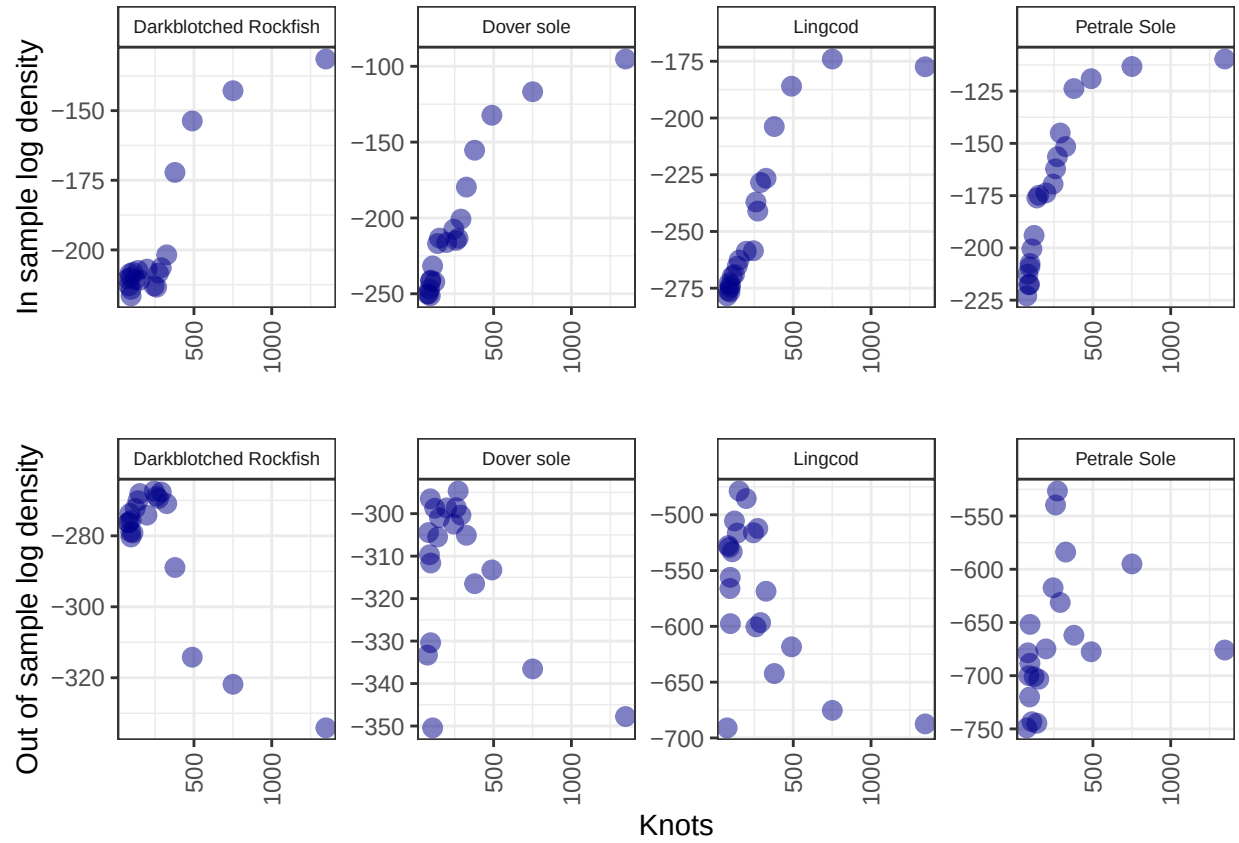


Figure 2: Binomial predictive densities for four species from the WCBTS survey, as a function of number of knots (using cutoff values from 3-120km)

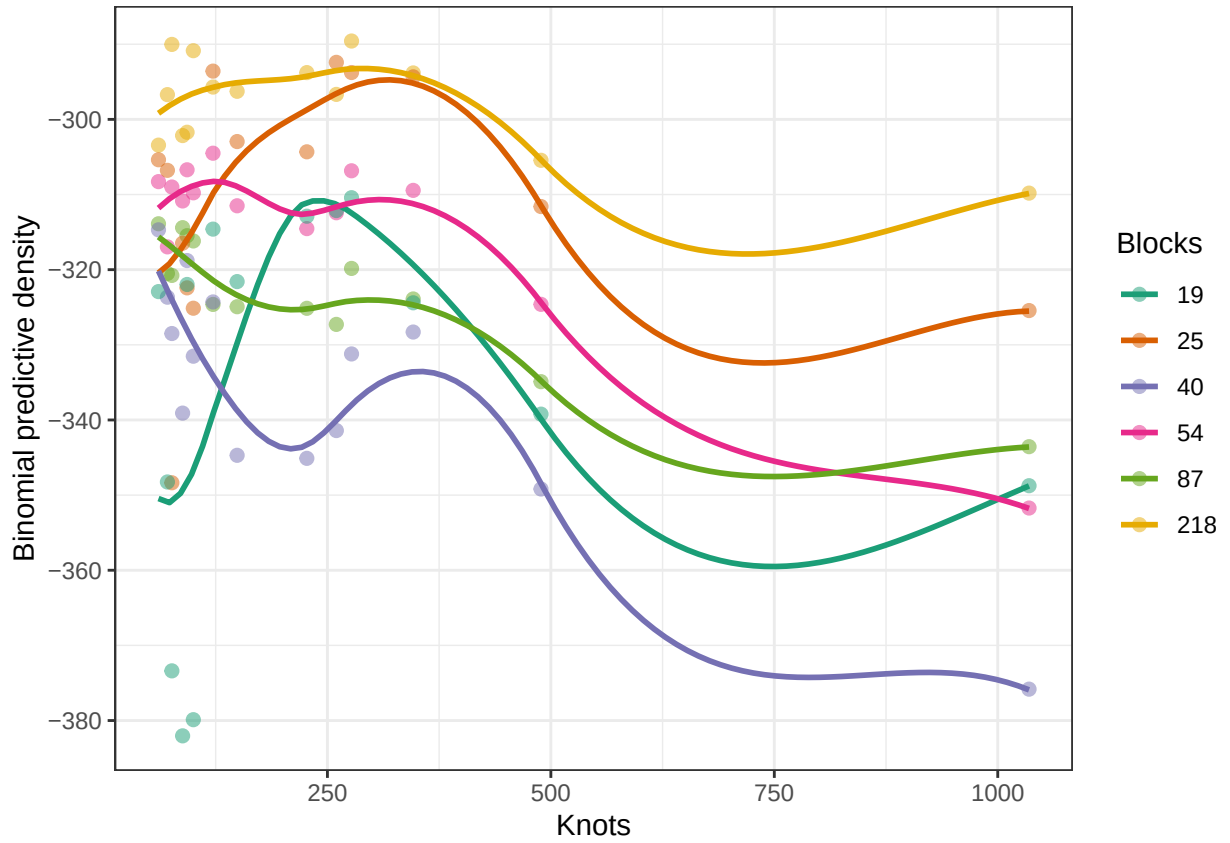


Figure 3: Binomial predictive density for Dover Sole, as a function of number of knots and number of spatial blocks used for 10-fold spatial cross-validation. Cutoff values between 5-135km are used across all scenarios.

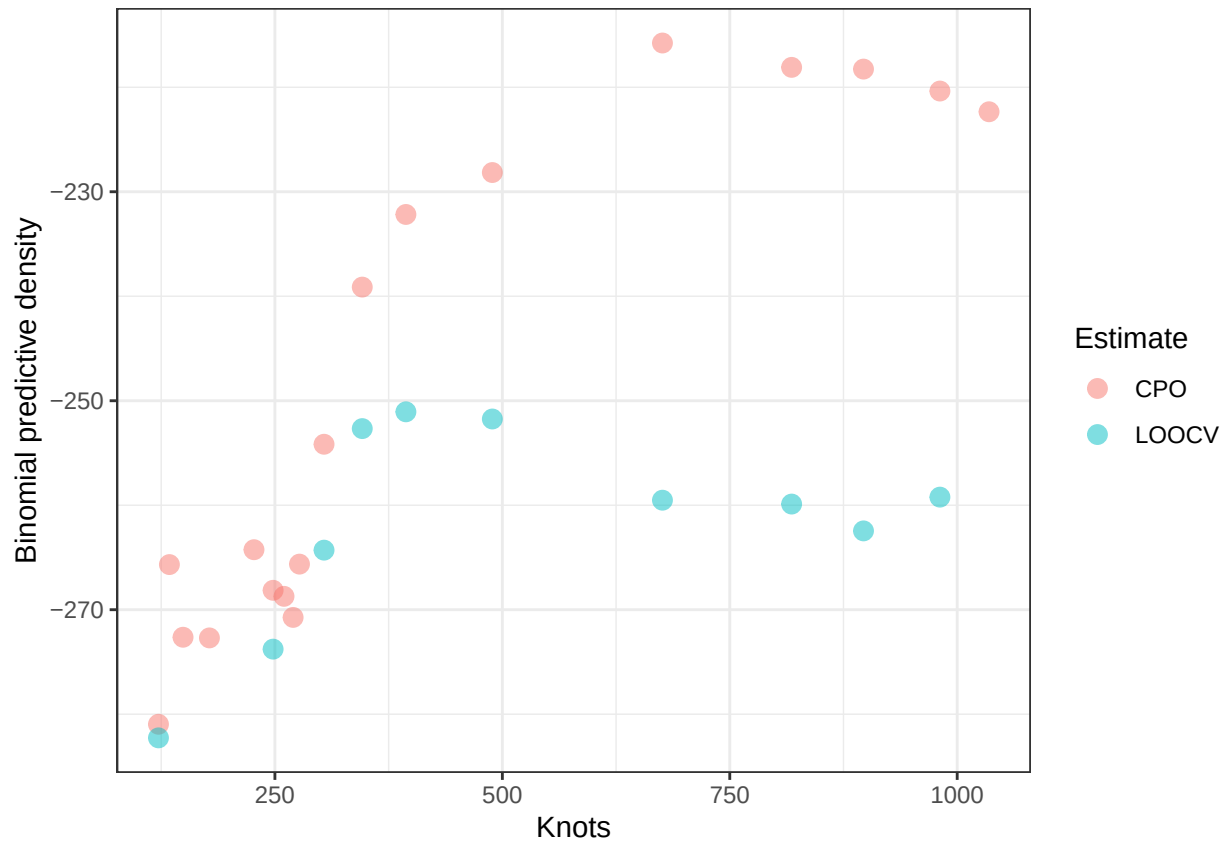


Figure 4: Comparison of CPO and LOOCV for the binomial presence-absence model of Dover sole. A wider range of cutoff values is shown for CPO (5-100km) than LOOCV (10-75km) to reduce computational time.

References

- Anderson, S.C., Keppel, E.A. & Edwards, A.M. (2019b). A reproducible data synopsis for over 100 species of British Columbia groundfish. *DFO Can. Sci. Advis. Sec. Res. Doc.*, **2019/041**.
- Anderson, S.C., Keppel, E.A. & Edwards, A.M. (2019a). A reproducible data synopsis for over 100 species of British Columbia groundfish. *DFO Can. Sci. Advis. Sec. Res. Doc.*, **2019/041**.
- Anderson, S.C. & Ward, E.J. (2019). Black swans in space: Modeling spatiotemporal processes with extremes. *Ecology*, **100**, e02403.
- Bachl, F.E., Lindgren, F., Borchers, D.L. & Illian, J.B. (2019). Inlabru: An R package for Bayesian spatial modelling from ecological survey data. *Methods in Ecology and Evolution*, **10**, 760–766.
- Bakka, H., Rue, H., Fuglstad, G.-A., Riebler, A., Bolin, D., Illian, J., Krainski, E., Simpson, D. & Lindgren, F. (2018). Spatial modeling with R-INLA: A review. *WIREs Computational Statistics*, **10**, e1443.
- Banerjee, A., Dunson, D.B. & Tokdar, S.T. (2012). Efficient Gaussian process regression for large datasets. *Biometrika*, **100**, 75–89.
- Banerjee, S. & Fuentes, M. (2012). Bayesian modeling for large spatial datasets. *WIREs Computational Statistics*, **4**, 59–66.
- Datta, A., Banerjee, S., Finley, A.O. & Gelfand, A.E. (2016). Hierarchical Nearest-Neighbor Gaussian Process Models for Large Geostatistical Datasets. *Journal of the American Statistical Association*, **111**, 800–812.
- Forney, K.A., Ferguson, M.C., Becker, E.A., Fiedler, P.C., Redfern, J.V., Barlow, J., Vilchis, I.L. & Ballance, L.T. (2012). Habitat-based spatial models of cetacean density in the eastern Pacific Ocean. *Endangered Species Research*, **16**, 113–133.
- Fuglstad, G.-A., Simpson, D., Lindgren, F. & Rue, H. (2018). Constructing priors that penalize the complexity of gaussian random fields. *Journal of the American Statistical Association*, **114**.
- Garton, N., Niemi, J. & Carriquiry, A. (2020). Knot selection in sparse gaussian processes with a variational objective function. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, **13**, 324–336.
- Godana, A.A., Mwalili, S.M. & Orwa, G.O. (2019). Dynamic spatiotemporal modeling of the infected rate of visceral leishmaniasis in human in an endemic area of Amhara regional state, Ethiopia. *PLOS ONE*, **14**, e0212934.

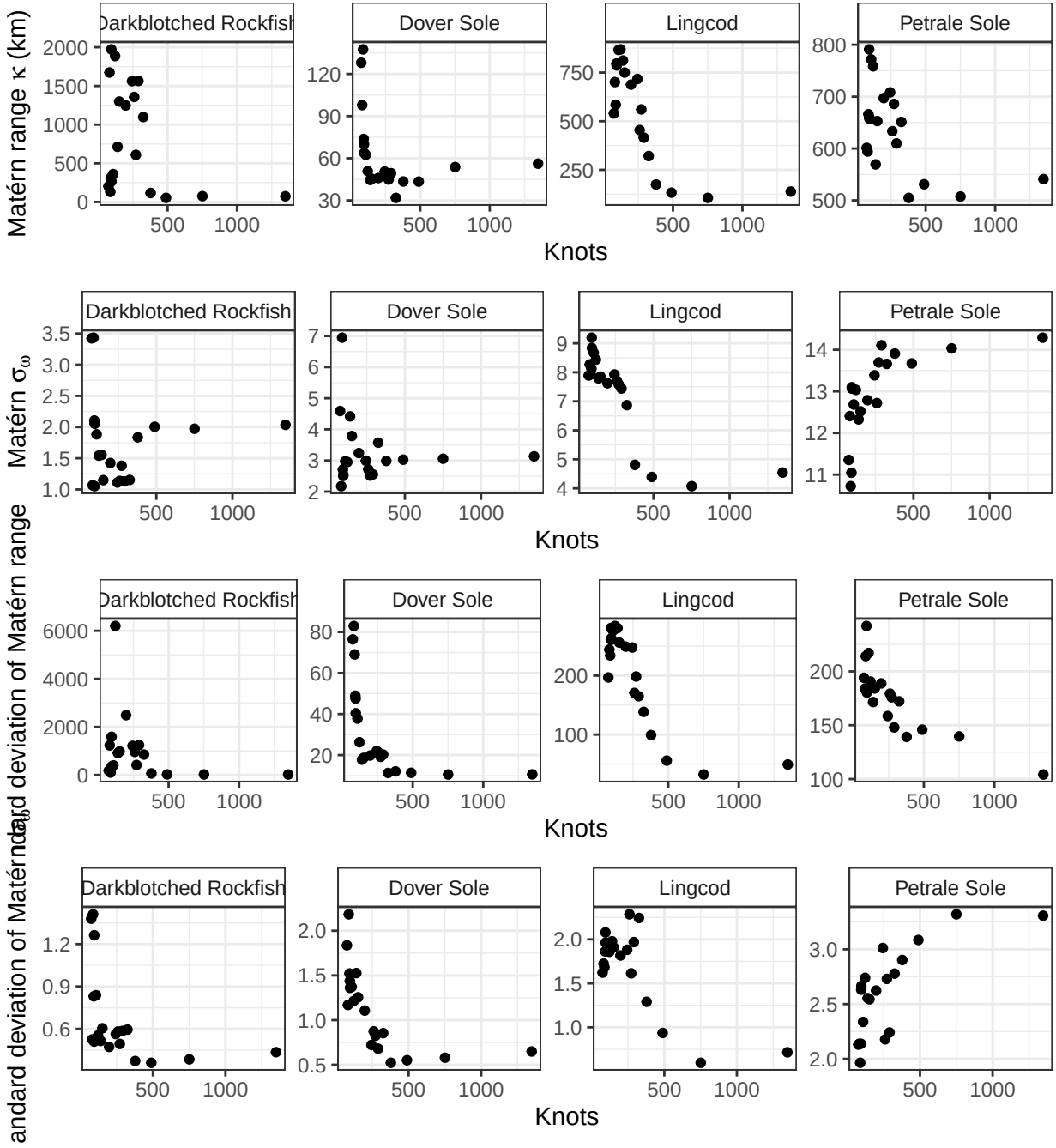


Figure 5: Estimates of the Matérn range and spatial scale, along with standard deviations for each parameter, across four species in the WCBTS survey. Cutoff values from 3-120km are shown.

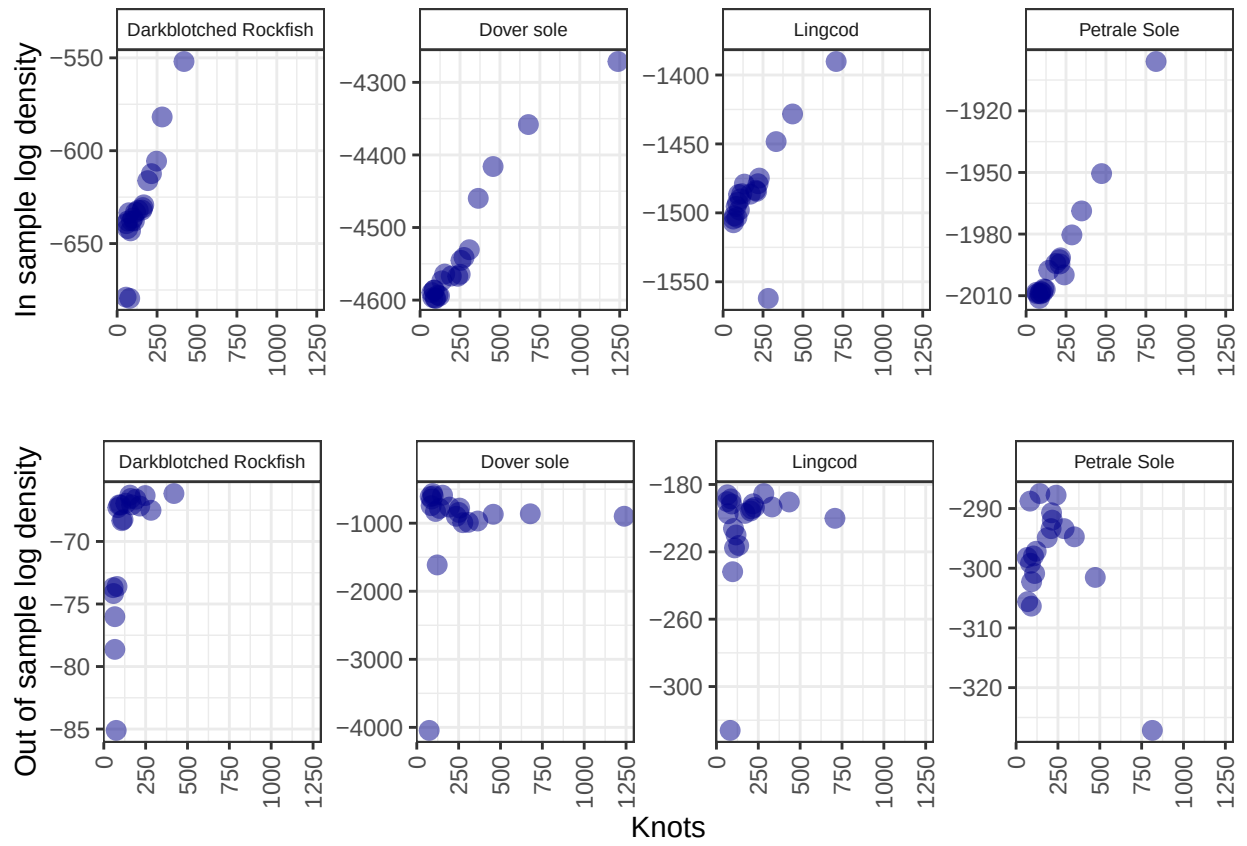


Figure 6: Gamma predictive densities for four species from the WCBTS survey, as a function of number of knots (using cutoff values from 3-120km)

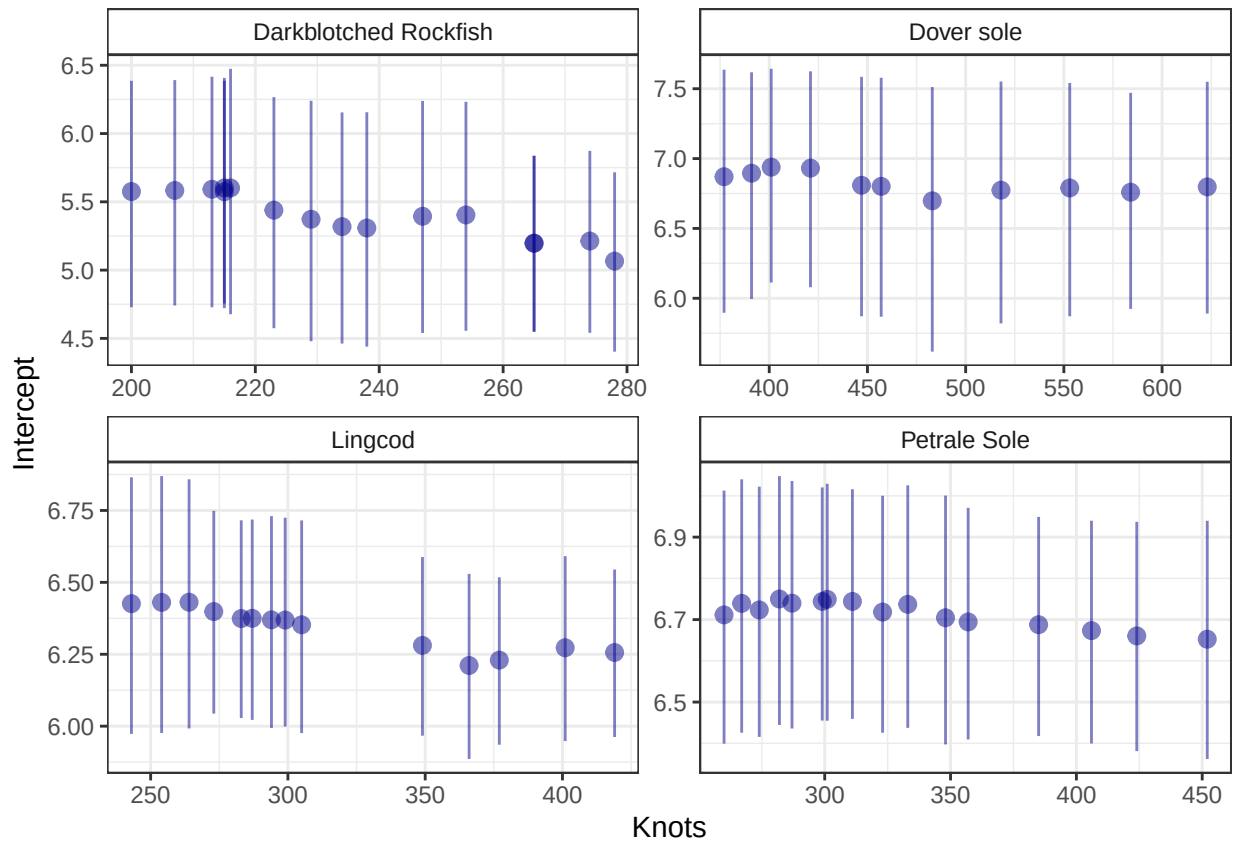


Figure 7: Global intercepts (and 95% CIs) for four species from the WCBTS survey, using the Gamma positive model, as a function of number of knots (using cutoff values from 3-120km)

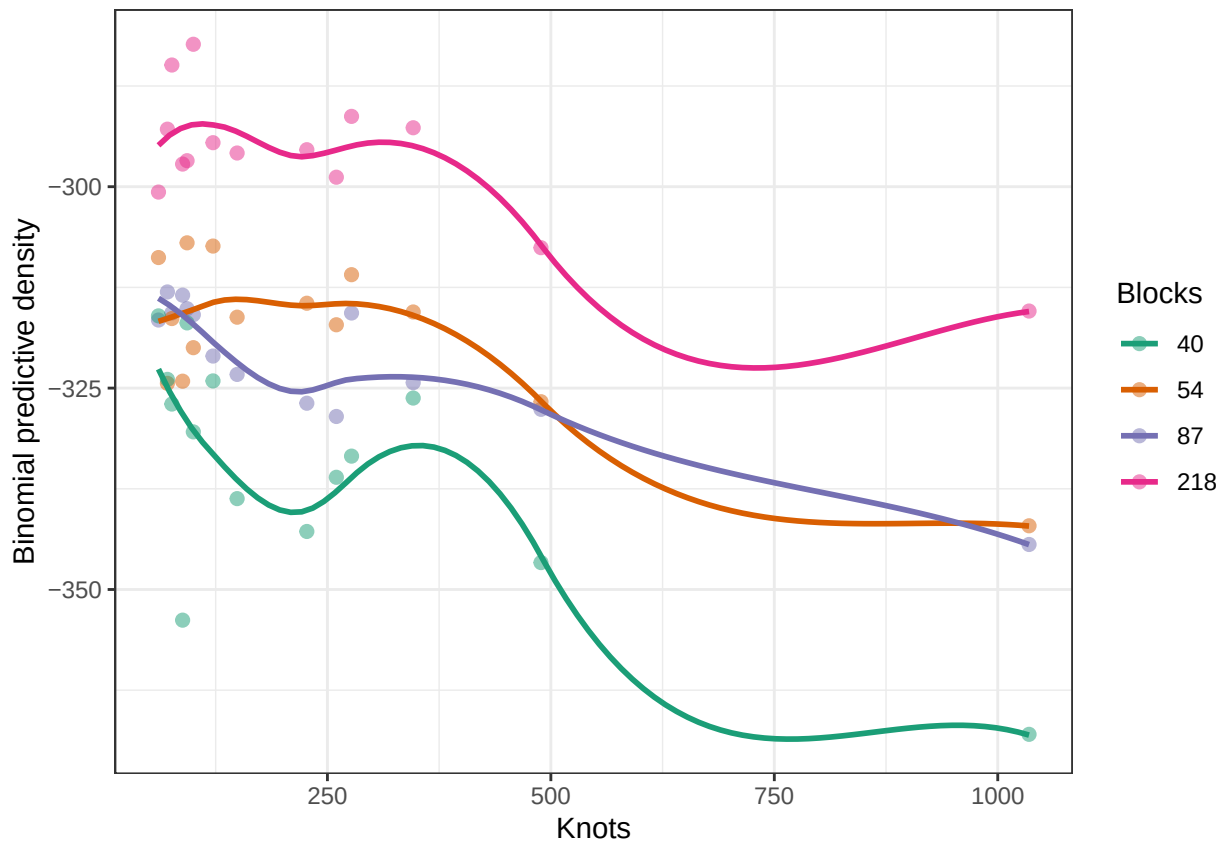


Figure 8: Binomial predictive density for Dover Sole, as a function of number of knots and number of spatial blocks used for 40-fold spatial cross-validation. Cutoff values between 5-135km are used across all scenarios.

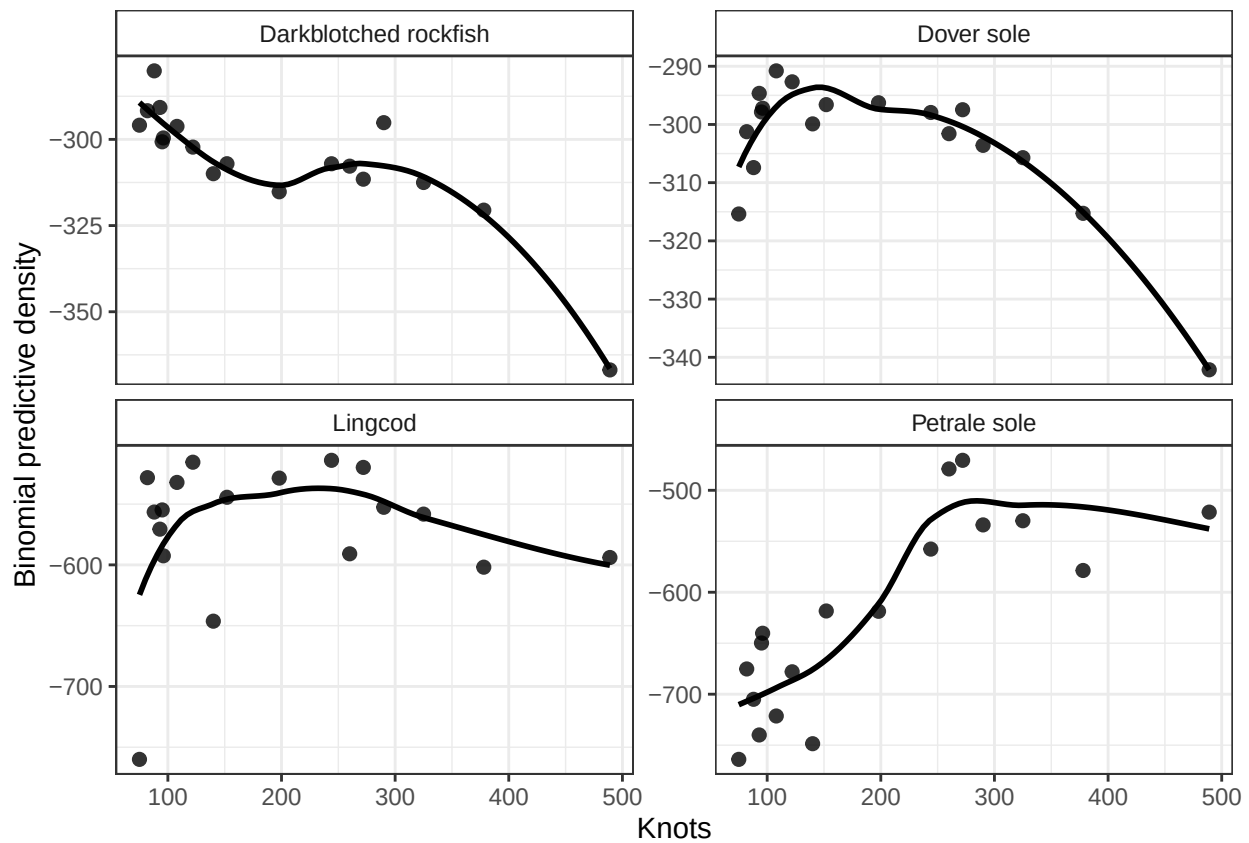


Figure 9: Binomial predictive densities for four species from the WCBTS survey, as a function of number of knots (using cutoff values from 3-120km). Estimation is done using Template Model Builder (TMB) instead of INLA.

Gómez-Rubio, V. *Bayesian inference with INLA*.

Held, L., Schrödle, B. & Rue, H. (2010). Posterior and Cross-validatory Predictive Checks: A Comparison of MCMC and INLA. *Statistical Modelling and Regression Structures: Festschrift in Honour of Ludwig Fahrmeir* (eds T. Kneib & G. Tutz), pp. 91–110. Physica-Verlag HD, Heidelberg.

Huang, J., Malone, B.P., Minasny, B., McBratney, A.B. & Triantafyllis, J. (2017). Evaluating a Bayesian modelling approach (INLA-SPDE) for environmental mapping. *The Science of the Total Environment*, **609**, 621–632.

Johnson, K.F., Thorson, J.T. & Punt, A.E. (2019). Investigating the value of including depth during spatiotemporal index standardization. *Fisheries Research*, **216**, 126–137.

Keller, A.A., Wallace, J.R. & Methot, R.D. (2017). *The Northwest Fisheries Science Center’s West Coast Groundfish Bottom Trawl Survey: History, Design, and Description*. Seattle, WA.

Kristensen, K., Nielsen, A., Berg, C.W., Skaug, H. & Bell, B.M. (2016). TMB: Automatic Differentiation and Laplace Approximation. *Journal of Statistical Software*, **70**, 1–21.

Latimer, A.M., Banerjee, S., Jr, H.S., Mosher, E.S. & Jr, J.A.S. (2009). Hierarchical models facilitate spatial analysis of large data sets: A case study on invasive plant species in the northeastern United States. *Ecology Letters*, **12**, 144–154.

Lindgren, F. & Rue, H. (2015). Bayesian Spatial Modelling with R-INLA. *Journal of Statistical Software*, **63**, 1–25.

Lindgren, F., Rue, H. & Lindström, J. (2011). An explicit link between Gaussian fields and Gaussian Markov random fields: The stochastic partial differential equation approach. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, **73**, 423–498.

Miller, D.L., Burt, M.L., Rexstad, E.A. & Thomas, L. (2013). Spatial models for distance sampling data: Recent developments and future directions. *Methods in Ecology and Evolution*, **4**, 1001–1010.

Miller, D.L., Glennie, R. & Seaton, A.E. (2020). Understanding the Stochastic Partial Differential Equation Approach to Smoothing. *Journal of Agricultural, Biological and Environmental Statistics*, **25**, 1–16.

Murase, H., Nagashima, H., Yonezaki, S., Matsukura, R. & Kitakado, T. (2009). Application of a generalized additive model (GAM) to reveal relationships between environmental factors and distributions of pelagic fish and krill: A case study in Sendai Bay, Japan. *ICES Journal of Marine Science*, **66**, 1417–1424.

- Mutanga, O., Adam, E. & Cho, M.A. (2012). High density biomass estimation for wetland vegetation using WorldView-2 imagery and random forest regression algorithm. *International Journal of Applied Earth Observation and Geoinformation*, **18**, 399–406.
- Osgood-Zimmerman, A. & Wakefield, J. (2021). A Statistical Introduction to Template Model Builder: A Flexible Tool for Spatial Modeling.
- Pettit, L.I. (1990). The conditional predictive ordinate for the normal distribution. *Journal of the Royal Statistical Society: Series B (Methodological)*, **52**, 175–184.
- R Core Team. (2021). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing, Vienna, Austria.
- Righetto, A.J., Faes, C., Vandendijck, Y. & Ribeiro, P.J. (2020). On the choice of the mesh for the analysis of geostatistical data using R-INLA. *Communications in Statistics - Theory and Methods*, **49**, 203–220.
- Ross, B.E., Hooten, M.B. & Koons, D.N. (2012). An Accessible Method for Implementing Hierarchical Models with Spatio-Temporal Abundance Data. *PLOS ONE*, **7**, e49395.
- Rue, H., Martino, S. & Chopin, N. (2009). Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, **71**, 319–392.
- Ruppert, D. (2002). Selecting the Number of Knots for Penalized Splines. *Journal of Computational and Graphical Statistics*, **11**, 735–757.
- Simpson, D.P., Rue, H., Martins, T.G., Riebler, A. & Sørbye, S.H. (2015). Penalising model component complexity: A principled, practical approach to constructing priors. *arXiv:1403.4630 [stat]*.
- Simpson, D., Rue, H., Riebler, A., Martins, T.G. & Sørbye, S.H. (2017). Penalising Model Component Complexity: A Principled, Practical Approach to Constructing Priors. *Statistical Science*, **32**, 1–28.
- Thorson, J.T. (2019a). Guidance for decisions using the Vector Autoregressive Spatio-Temporal (VAST) package in stock, ecosystem, habitat and climate assessments. *Fisheries Research*, **210**, 143–161.
- Thorson, J.T. (2019b). Measuring the impact of oceanographic indices on species distribution shifts: The spatially varying effect of cold-pool extent in the eastern Bering Sea. *Limnology and Oceanography*, **64**, 2632–2645.
- Thorson, J.T. & Barnett, L.A.K. (2017). Comparing estimates of abundance trends and distribution shifts using single- and multispecies models of fishes and biogenic habitat. *ICES Journal of Marine Science*,

74, 1311–1321.

- Tolimieri, N., Wallace, J. & Haltuch, M. (2020). Spatio-temporal patterns in juvenile habitat for 13 groundfishes in the California Current Ecosystem. *PLOS ONE*, **15**, e0237996.
- Valavi, R., Elith, J., Lahoz-Monfort, J.J. & Guillera-Arroita, G. (2019). blockCV: An r package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution*, **10**, 225–232.
- Williamson, L.D., Scott, B.E., Laxton, M.R., Bachl, F.E., Illian, J.B., Brookes, K.L. & Thompson, P.M. Spatiotemporal variation in harbor porpoise distribution and foraging across a landscape of fear. *Marine Mammal Science*, **n/a**.
- Wilson, B. (2021). Evaluating the INLA-SPDE approach for Bayesian modeling of earthquake damages from geolocated cluster data.
- Wood, S.N. (2003). Thin plate regression splines. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, **65**, 95–114.
- Yue, Y.R., Simpson, D., Lindgren, F. & Rue, H. (2014). Bayesian adaptive smoothing splines using stochastic differential equations. *Bayesian Analysis*, **9**, 397–424.